

E-MOSFET AS A DIGITAL SWITCH

E-MOSFET APPLICATION

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TOPIC OUTLINE

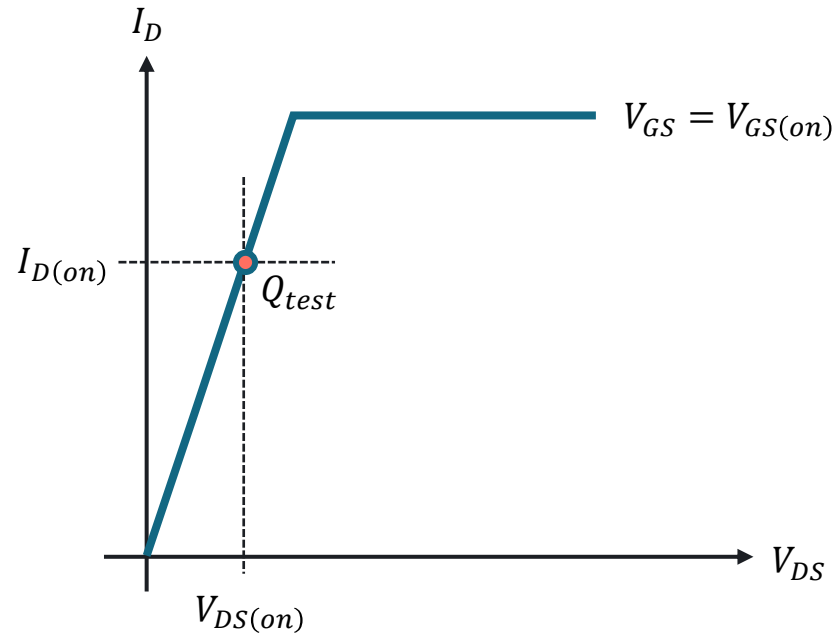
The Ohmic Region

Active-load Switching



OHMIC REGION

Measuring $R_{DS(on)}$



Drain-Source ON Resistance

$$R_{DS(on)} = \frac{V_{DS(on)}}{I_{D(on)}}$$

For instance, at the test point, a VN2406L has $V_{DS(on)} = 1V$ and $I_{D(on)} = 100mA$.

$$R_{DS(on)} = \frac{1V}{100\text{ mA}}$$

$$R_{DS(on)} = 10\Omega$$

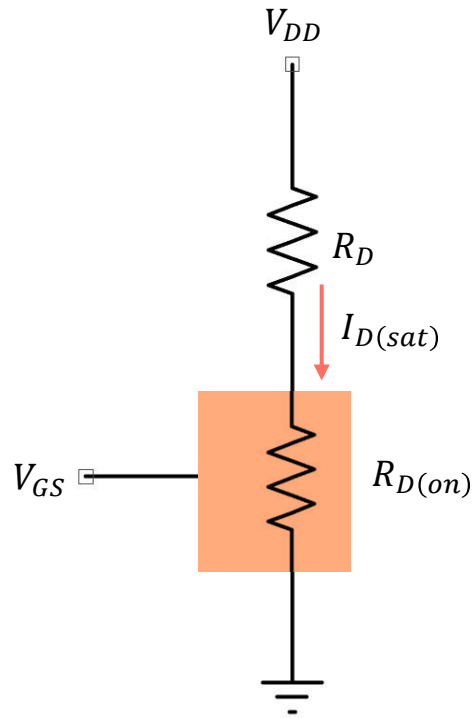


SMALL-SIGNAL E-MOSFETs

Device	$V_{GS(th)}$	$V_{GS(on)}$	$I_{D(on)}$	$R_{DS(on)}$	$I_{D(max)}$	$P_{D(max)}$
VN2406L	1.5V	2.5V	100 mA	10 Ω	200mA	350mW
BS107	1.75V	2.6V	20 mA	28 Ω	250mA	350mW
2N7000	2V	4.5V	75 mA	6 Ω	200mA	350mW
VN10LM	2.5V	5V	200 mA	7.5 Ω	300mA	1W
MPF930	2.5V	10V	1 A	0.9 Ω	2A	1W
IRFD120	3V	10V	600 mA	0.3 Ω	1.3A	1W

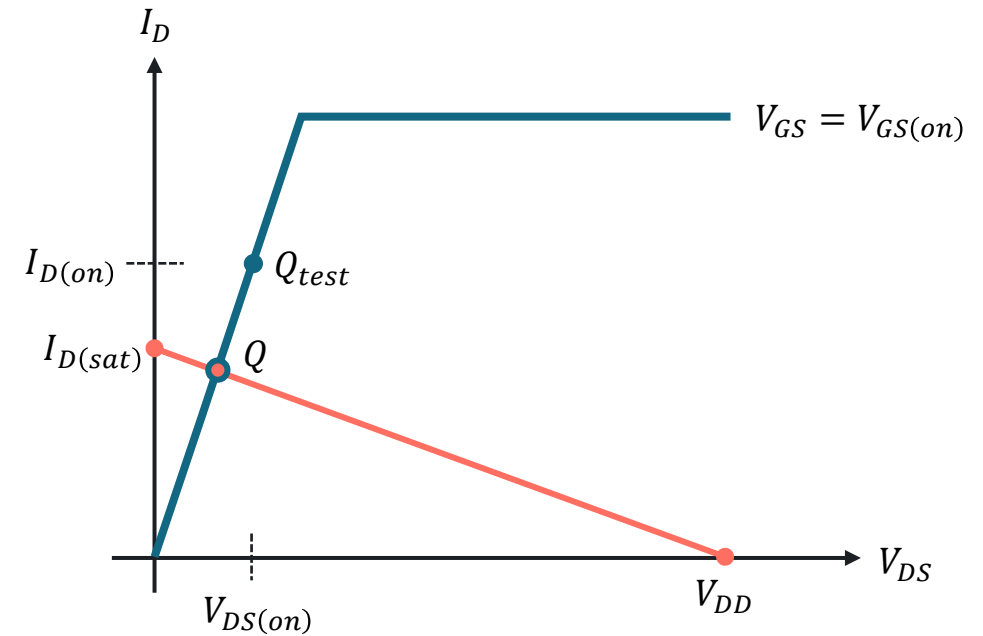


BIASING IN OHMIC REGION



Drain saturation current

$$i_{D(sat)} = \frac{V_{DD}}{R_D}$$

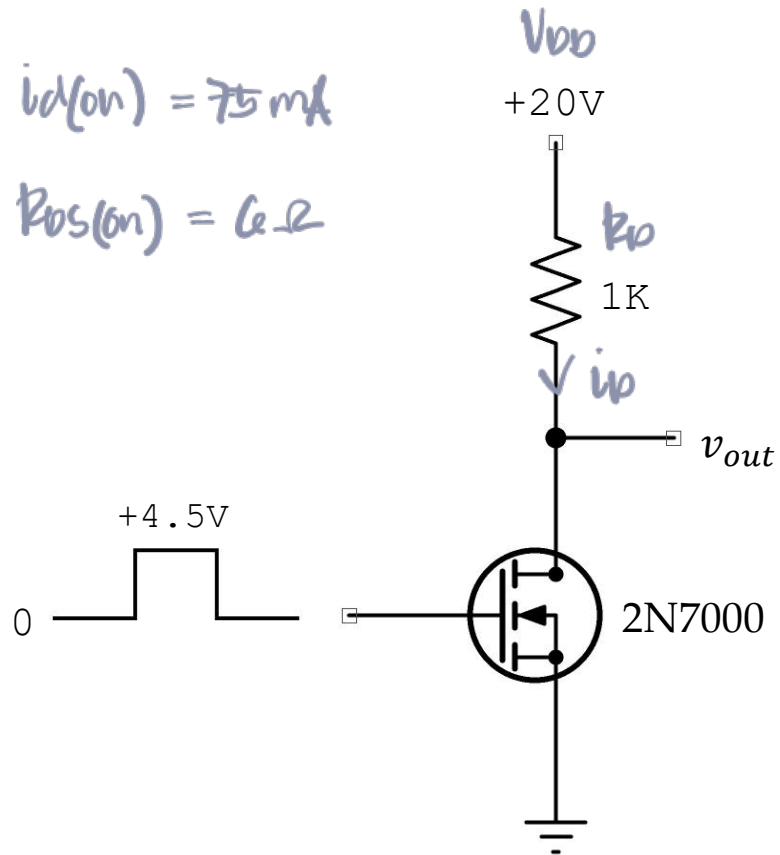


$I_{D(sat)}$ less than $I_{D(on)}$ with $V_{GS} = V_{GS(on)}$ ensures saturation.



EXERCISE

What is the output voltage in the given circuit?



Solution

Check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

$$i_{D(SAT)} = \frac{20}{1K}$$

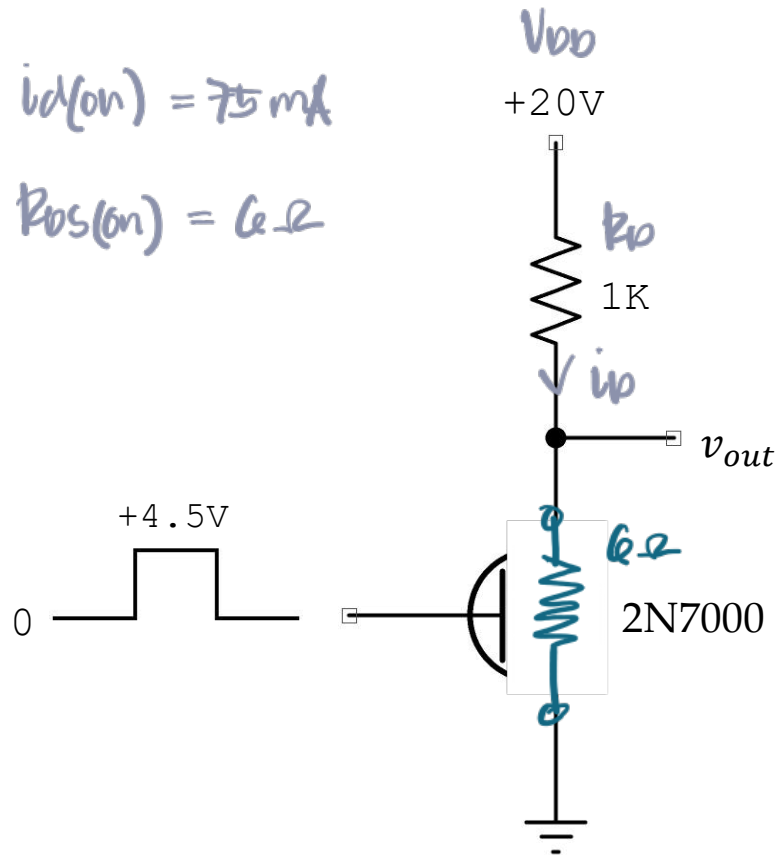
$$i_{D(SAT)} = 20 \text{ mA} < 75 \text{ mA}$$

$$\therefore R_{DS(on)} = 6 \Omega$$



EXERCISE

What is the output voltage in the given circuit?



Solution

Let $V_{GS} = 4.5V$

$$V_{out} = V_{DD} \frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$V_{out} = 20 \frac{6}{1k + 6}$$

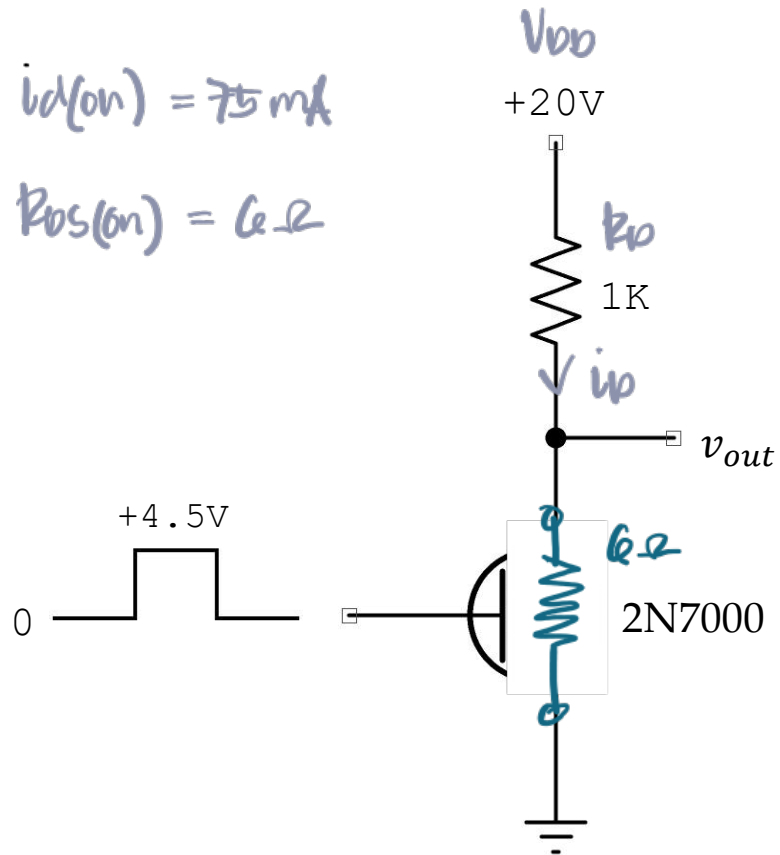
$$V_{out} = 0.12V$$

ans



EXERCISE

What is the output voltage in the given circuit?



Solution

Let $v_{GS} = \text{LOW}$

$$v_{out} = V_{DD} \frac{\cancel{R_{DS(on)}}}{R_D + \cancel{R_{DS(on)}}}$$

$$v_{out} = V_{DD}$$

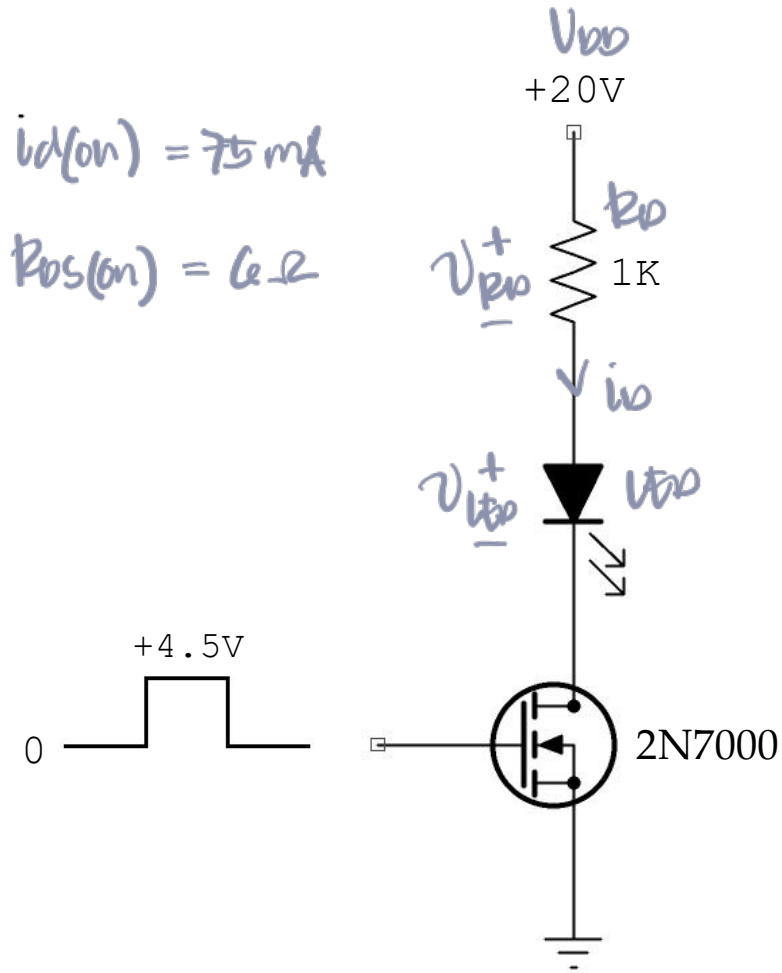
$$v_{out} = 20V$$

correct



EXERCISE

Calculate the LED current in the given circuit.



Solution

Check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

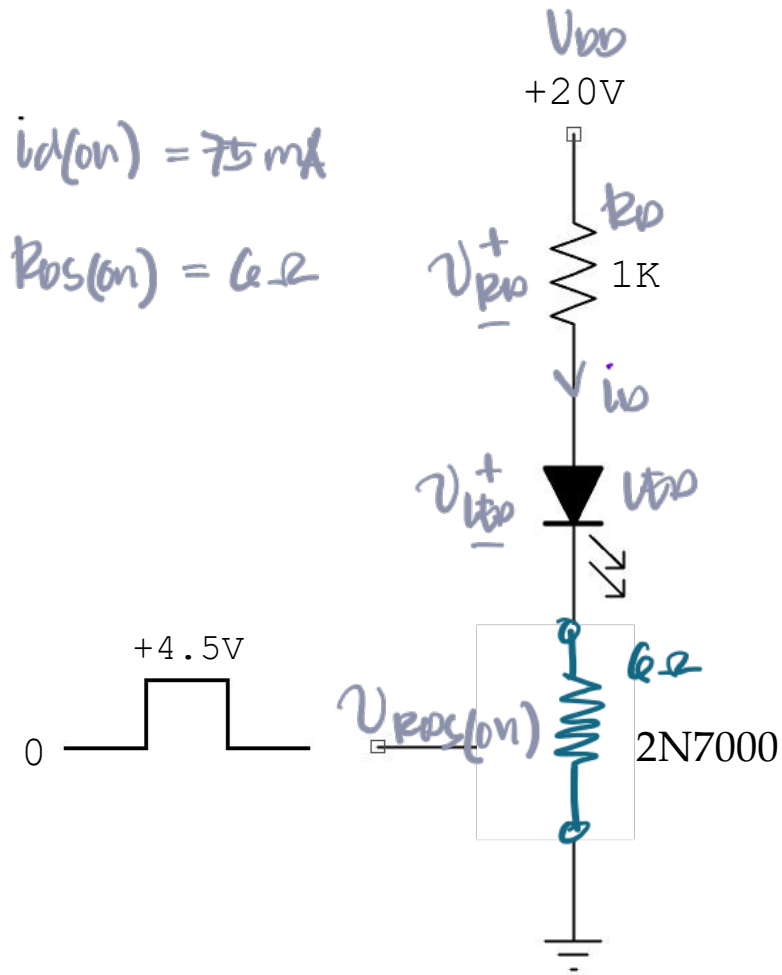
$$i_{D(SAT)} = \frac{20}{1K}$$

$$i_{D(SAT)} = 20 \text{ mA} < 75 \text{ mA}$$

$$\therefore R_{DS(on)} = 6 \Omega$$

EXERCISE

Calculate the LED current in the given circuit.



Solution

KVL @ D-S

$$-V_{DD} + V_{R_D} + V_{LED} + V_{R_{DS(\text{on})}} = 0$$

$$V_{R_D} + V_{R_{DS(\text{on})}} = V_{DD} - V_{LED}$$

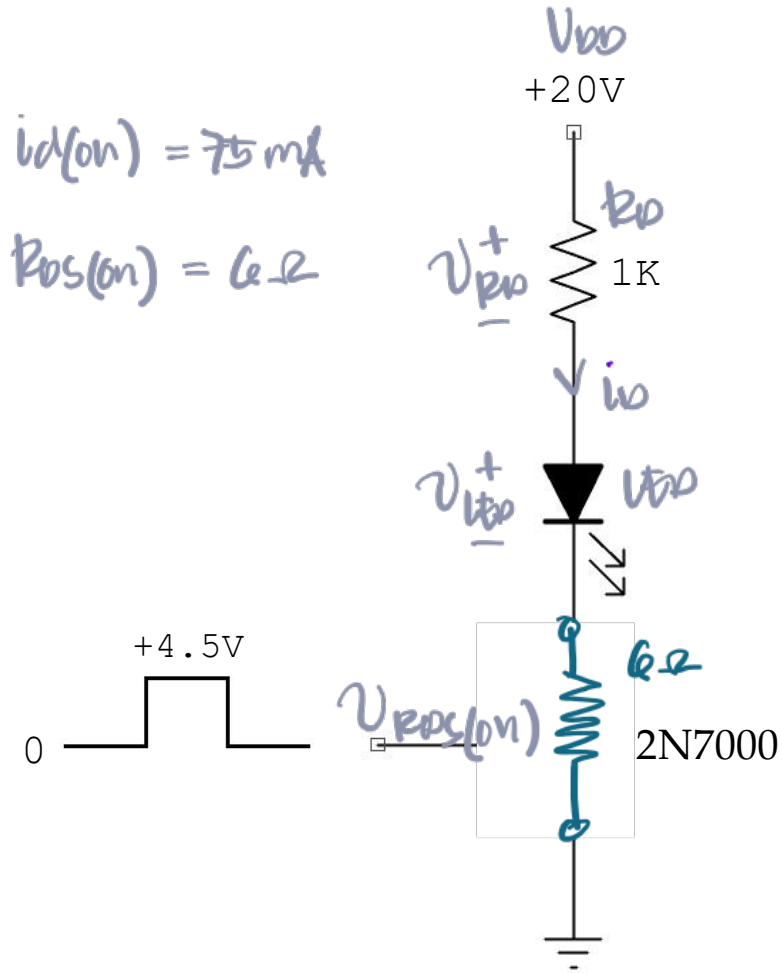
$$i_D R_D + i_D R_{DS(\text{on})} = V_{DD} - V_{LED}$$

$$\frac{i_D (R_D + R_{DS(\text{on})})}{R_D + R_{DS(\text{on})}} = \frac{V_{DD} - V_{LED}}{R_D + R_{DS(\text{on})}}$$

Let 2V

EXERCISE

Calculate the LED current in the given circuit.



Solution

KVL @ D-S

$$i_D = \frac{20 - 2}{1K + 6}$$

$$i_D = 17.89 \text{ mA}$$

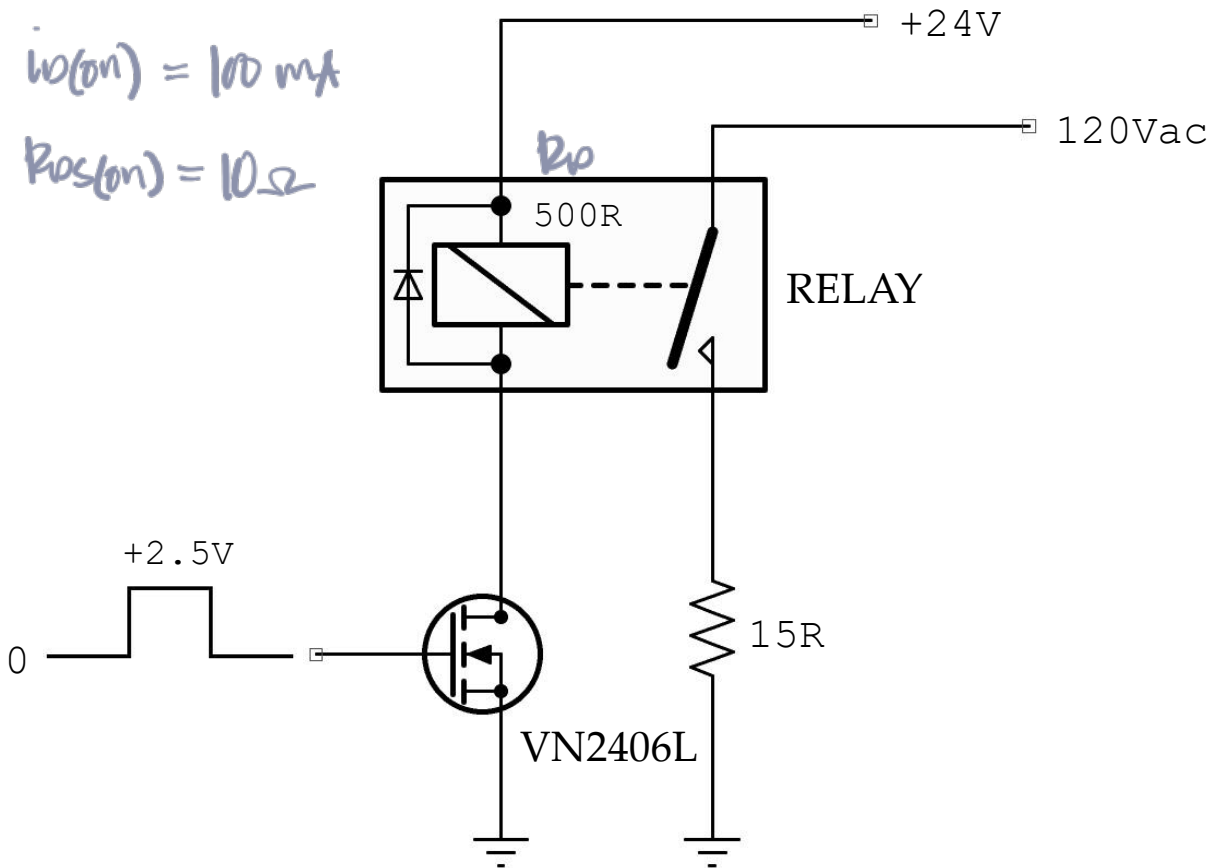
ans

note : This is true if $V_{GS} = \text{HIGH}$.

If $V_{GS} = \text{LOW}$, $i_D = 0$.

EXERCISE

What does the given circuit do if a coil current of 30mA or more closes the relay contact?



Solution

check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

$$i_{D(SAT)} = \frac{24}{500}$$

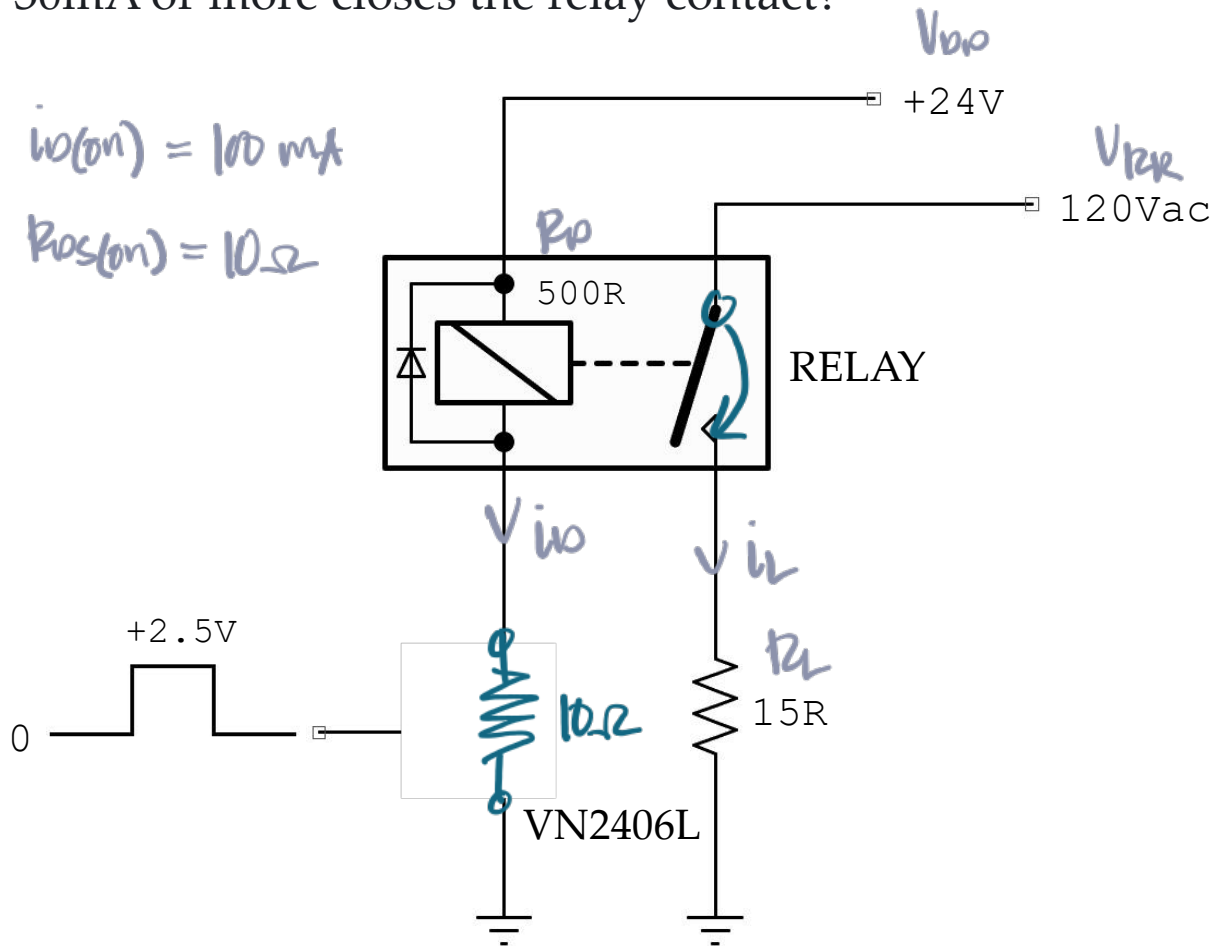
$$i_{D(SAT)} = 48 \text{ mA} < 100 \text{ mA}$$

$$\therefore R_{DS(on)} = 10 \Omega$$



EXERCISE

What does the given circuit do if a coil current of 30mA or more closes the relay contact?



Solution

Drain Current

$$i_o = \frac{V_{o0}}{R_o + R_{DS(on)}}$$

$$i_o = \frac{24}{570 + 10}$$

$$i_{w0} = 47.06 \text{ mA}$$

Load Current

$$i_L = \frac{V_{R_F}}{R_L}$$

$$= \frac{120}{15}$$

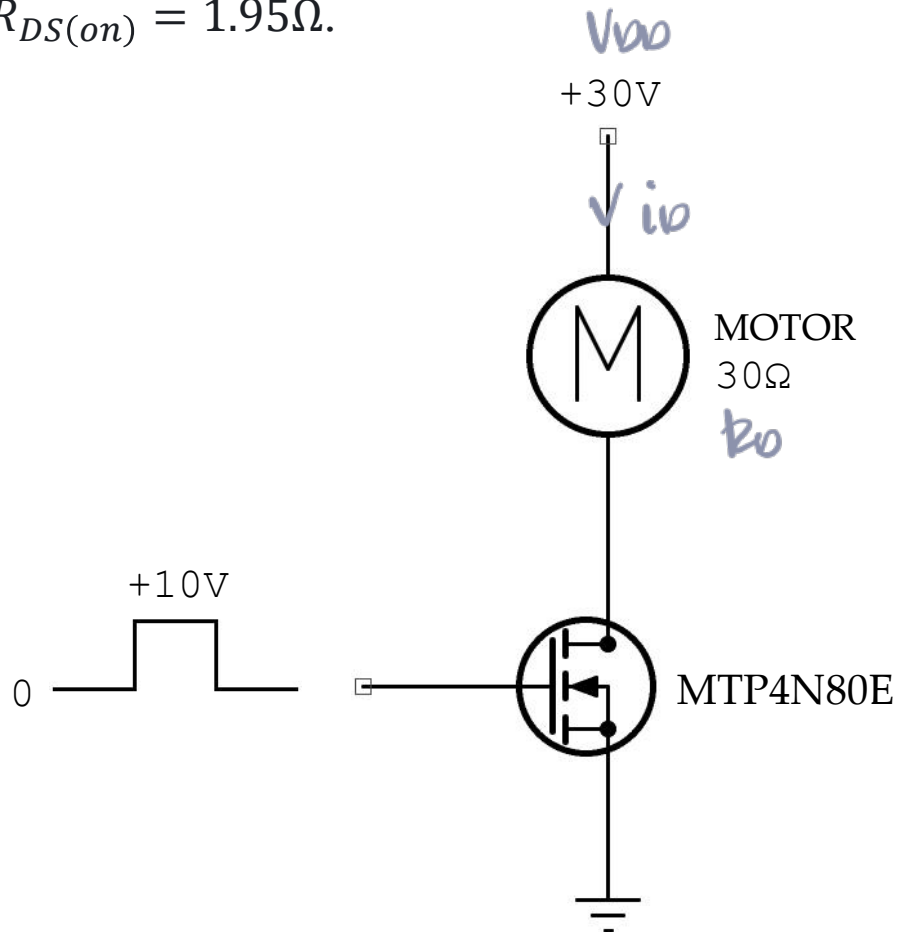
$$i_L = 8 \text{ mA}$$

Ans



EXERCISE

What is the current through the motor winding? For an MTP4N80E: $V_{GS(on)} = 10\text{ V}$, $I_{D(on)} = 2\text{ A}$, and $R_{DS(on)} = 1.95\Omega$.



Solution

check if in saturation

$$i_{D(SAT)} = \frac{V_{DD}}{R_D}$$

$$i_{D(SAT)} = \frac{30}{30}$$

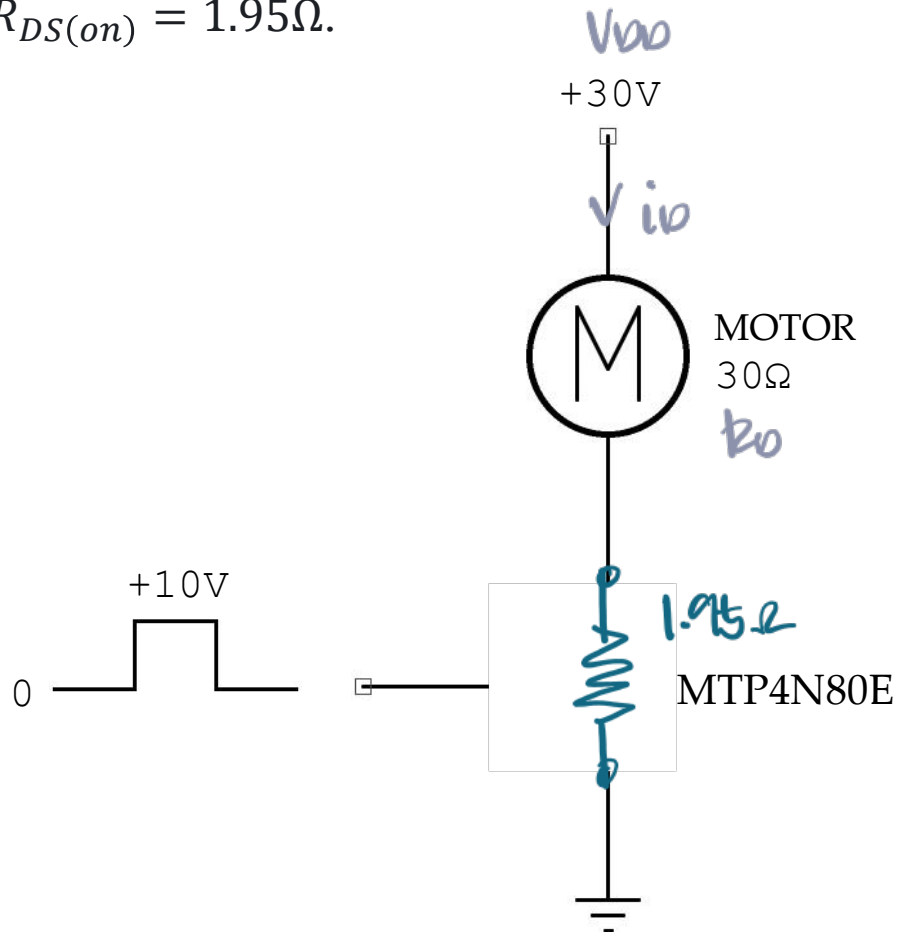
$$i_{D(SAT)} = 1\text{ A} < i_{D(on)} = 2\text{ A}$$

$$\therefore R_{DS(on)} = 1.95\Omega$$



EXERCISE

What is the current through the motor winding? For an MTP4N80E: $V_{GS(on)} = 10\text{ V}$, $I_{D(on)} = 2\text{ A}$, and $R_{DS(on)} = 1.95\Omega$.



Solution

Drain Current

$$i_D = \frac{V_{DD}}{R_D + R_{DS(on)}}$$

$$i_D = \frac{30}{30 + 1.95}$$

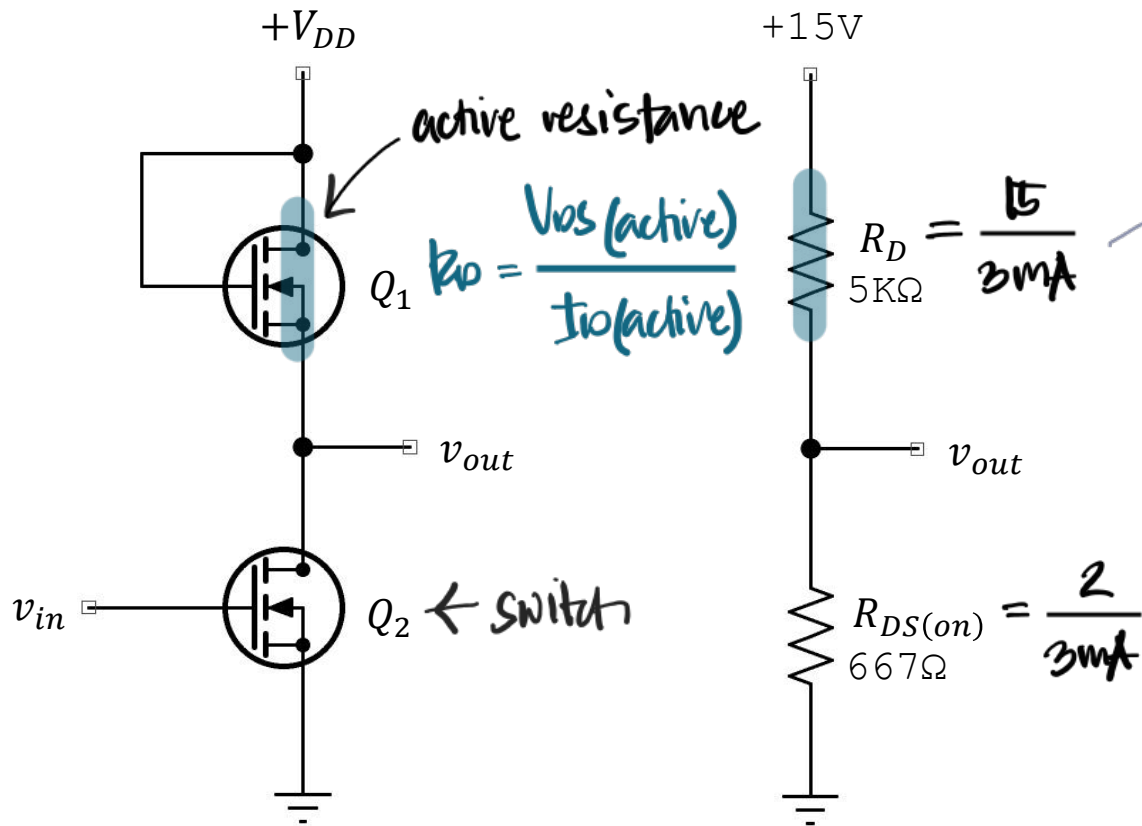
$$i_D = 938.97\text{ mA}$$

ans

ACTIVE-LOAD SWITCHING



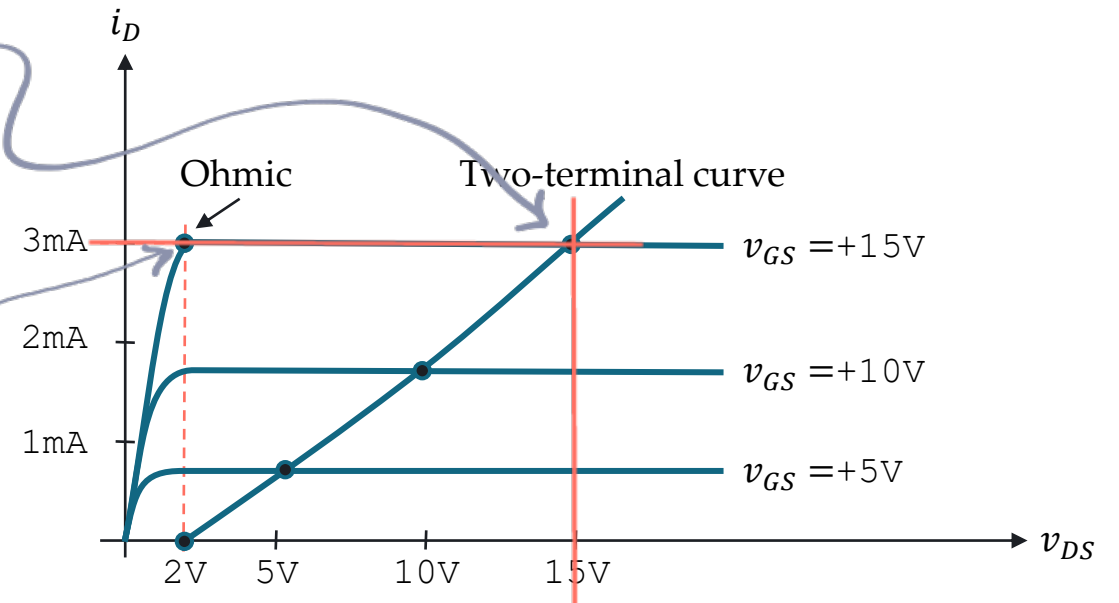
ACTIVE-LOAD SWITCHING



Active Load

Equivalent Circuit

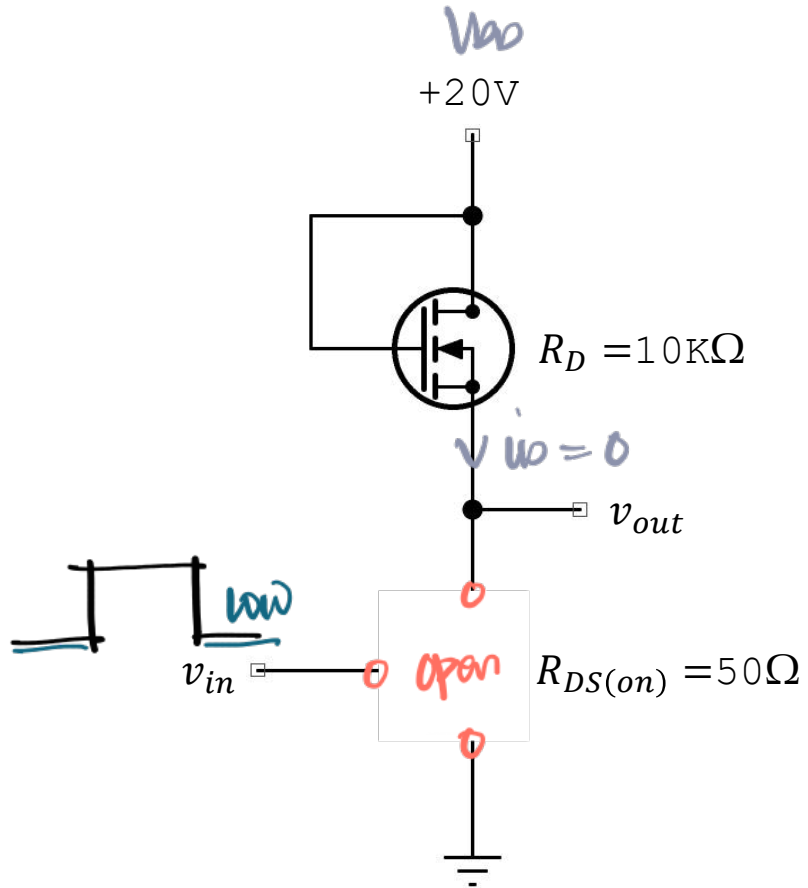
$V_{GS} = V_{DS}$ produces a two-terminal curve



EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When $v_{in} = LOW$

$$v_{out} = 20V$$

ans

Output is **pull-up** to the supply voltage.

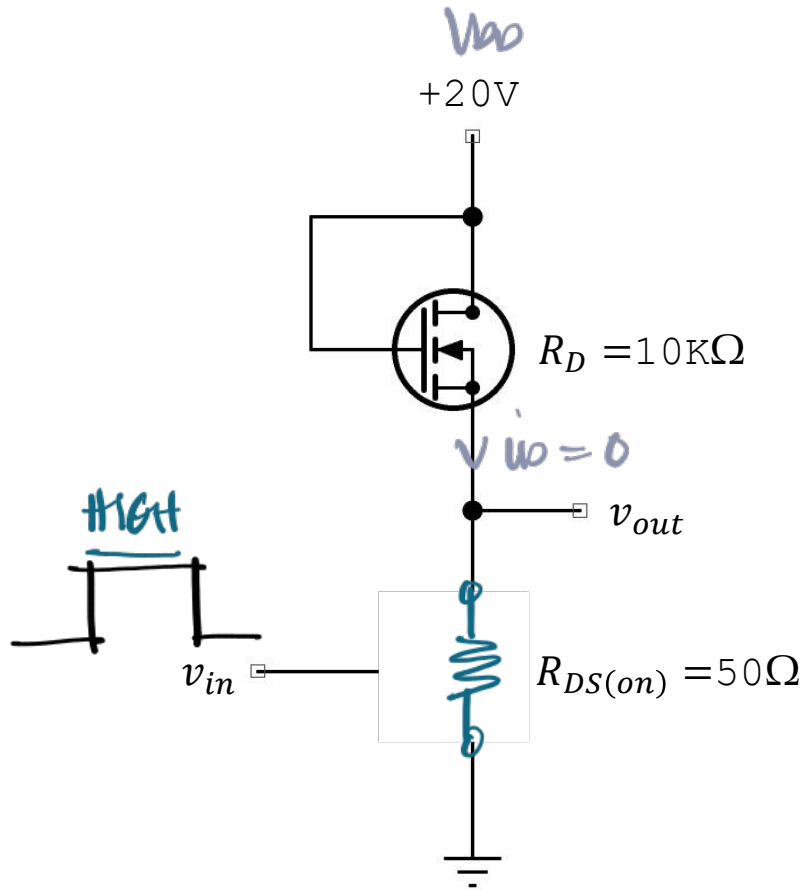
input = low
output = high } inverter



EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When $v_{in} = \text{HIGH}$

$$v_{out} = V_{DD} \frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$v_{out} = 20 \frac{50}{10K + 50}$$

$$v_{out} = 99.5 \text{ mV}$$

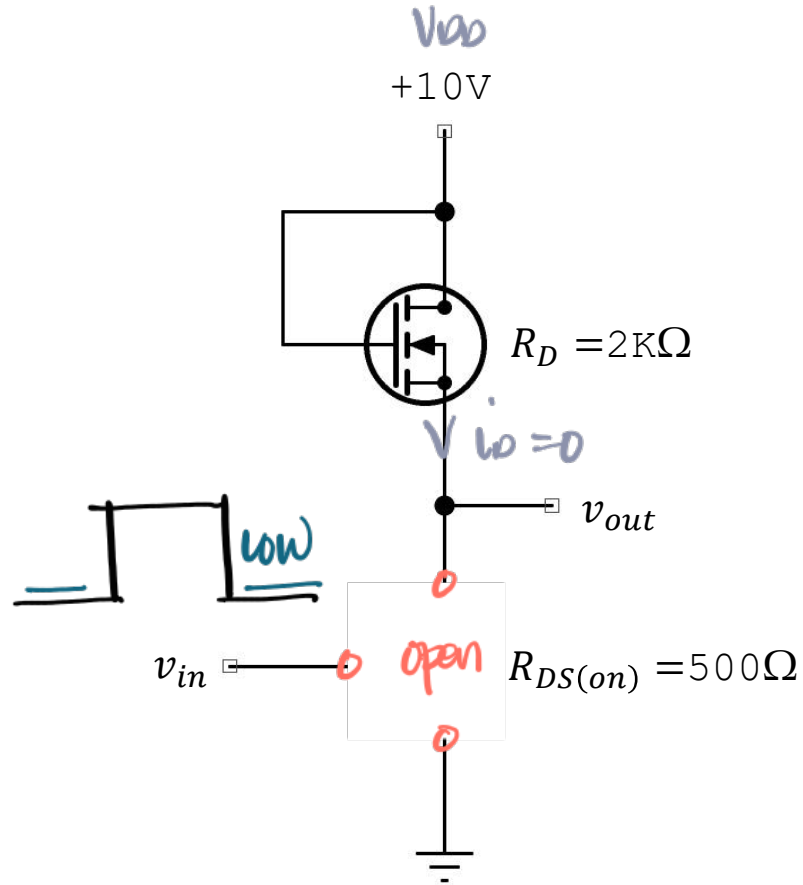
ans

input = HIGH } inverter
output = LOW

EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When $v_{in} = \text{low}$

$$v_{out} = 10V$$

ans

Output is **pull-up** to the supply voltage

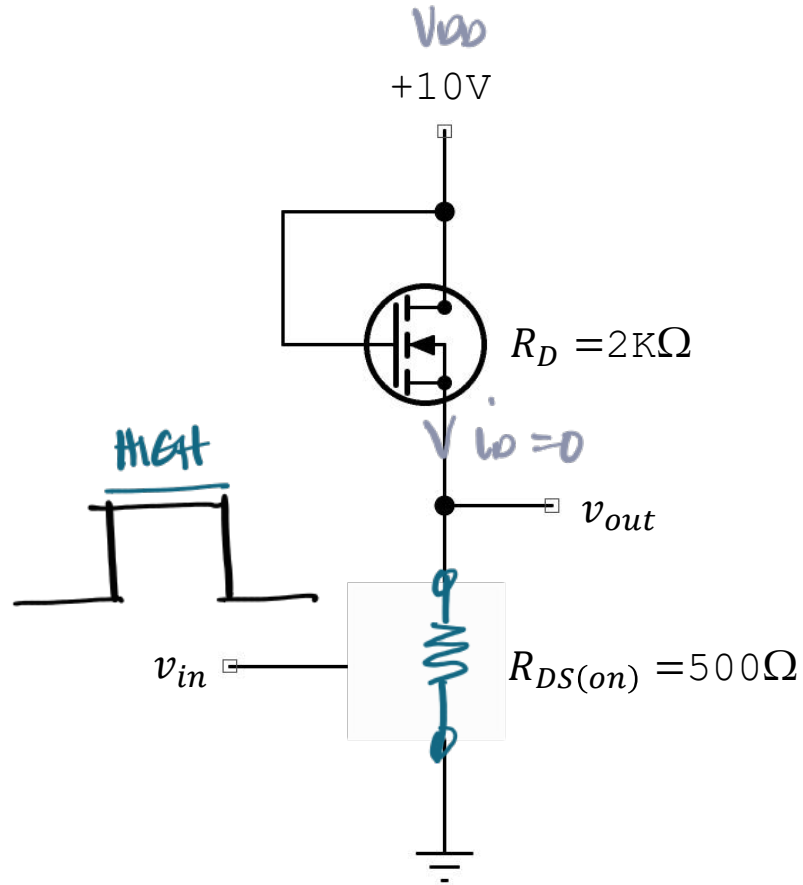
input = low
output = high } inverter



EXERCISE

What is the output voltage when the input is low?

When the input is high?



Solution

When $v_{in} = \text{HIGH}$

$$v_{out} = V_{DD} \frac{R_{DS(on)}}{R_D + R_{DS(on)}}$$

$$v_{out} = 10 \frac{500}{2k + 500}$$

$$v_{out} = 2V$$

ans



LABORATORY

